

CELEBAL EXCELLENCE INTERNSHIP PROGRAM

DATA ENGINEERING

WEEK - 2 ASSIGNMENT REPORT - I

KAGGLE DATASET ASSIGNMENT

SUBMITTED BY:-

ANUSHKA SHARMA

(JKLU)

STEP 1. Load Dataset

The Superstore dataset was downloaded from Kaggle and loaded into a MySQL database. Python was used to read the CSV file and insert the records into the superstore table in MySQL.

STEP 2. Data Exploration

SQL Query :-

```
DESCRIBE superstore;
```

Output :-

Result Grid  Filter Rows: Export:  Wrap Cell Content: 						
	Field	Type	Null	Key	Default	Extra
▶	Row_ID	int	YES		NULL	
	Order_ID	varchar(30)	YES		NULL	
	Order_Date	varchar(20)	YES		NULL	
	Ship_Date	varchar(20)	YES		NULL	
	Ship_Mode	varchar(50)	YES		NULL	
	Customer_ID	varchar(30)	YES		NULL	
	Customer_Name	varchar(100)	YES		NULL	
	Segment	varchar(50)	YES		NULL	
	Country	varchar(50)	YES		NULL	
	City	varchar(50)	YES		NULL	
	State	varchar(50)	YES		NULL	
	Postal_Code	int	YES		NULL	
	Region	varchar(50)	YES		NULL	
	Product_ID	varchar(50)	YES		NULL	
	Category	varchar(50)	YES		NULL	
	Sub_Category	varchar(50)	YES		NULL	
	Product Name	varchar(255)	YES		NULL	
	Sales	decimal(12,4)	YES		NULL	
	Quantity	int	YES		NULL	
	Discount	decimal(12,4)	YES		NULL	
	Profit	decimal(12,4)	YES		NULL	

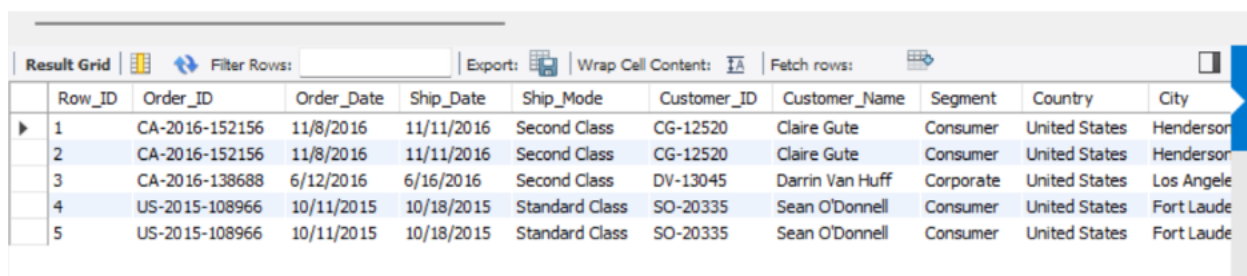
Result 9 ×

Insight :- This query shows the columns and data types present in the dataset. It helps in understanding the structure of the table before analysis.

SQL Query :-

```
SELECT *  
FROM superstore  
LIMIT 5;
```

Output :-



	Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_Name	Segment	Country	City
▶	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson
	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson
	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles
	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale
	5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale

Insight :- The first few records give an overview of the available data. It helps verify that the dataset has been imported correctly.

STEP 3. WHERE Filters

SQL Query :-

```
-- data of west region  
SELECT *  
FROM superstore  
WHERE Region = 'West';
```

Output :-

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Fetch rows:

	stomer_ID	Customer_Name	Segment	Country	City	State	Postal_Code	Region	Product_ID	Category
▶	13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000240	Office Supp
	11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-10001487	Furniture
	11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-10002833	Office Supp
	11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002275	Technology
	11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-BI-10003910	Office Supp
	11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AP-10002892	Office Supp
	11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-TA-10001539	Furniture
	11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002033	Technology
	15070	Irene Maddox	Consumer	United States	Seattle	Washington	98103	West	OFF-BI-10003656	Office Supp

superstore 14

Insight :- This query shows all orders from the West region. It helps analyze sales in a specific geographical area.

SQL Query :-

```
-- data of office supplies category
SELECT *
FROM superstore
WHERE Category = 'Office Supplies';
```

Output :-

Result Grid										Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
Segment	Country	City	State	Postal_Code	Region	Product_ID	Category	Sub_Category	Product_Name				
Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels				
Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Roll Cart Storage				
Consumer	United States	Los Angeles	California	90032	West	OFF-AR-10002833	Office Supplies	Art	Newell 322				
Consumer	United States	Los Angeles	California	90032	West	OFF-BI-10003910	Office Supplies	Binders	DXL Angle-View Binders				
Consumer	United States	Los Angeles	California	90032	West	OFF-AP-10002892	Office Supplies	Appliances	Belkin F5C206VTEL 6 Outlets				
Consumer	United States	Concord	North Carolina	28027	South	OFF-PA-10002365	Office Supplies	Paper	Xerox 1967				
Consumer	United States	Seattle	Washington	98103	West	OFF-BI-10003656	Office Supplies	Binders	Fellowes PB200 Plastic Cover				
Home Office	United States	Fort Worth	Texas	76106	Central	OFF-AP-10002311	Office Supplies	Appliances	Holmes Replacement Filter				
Home Office	United States	Fort Worth	Texas	76106	Central	OFF-BI-10000756	Office Supplies	Binders	Storex DuraTech Recycled				

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Insight :- This query displays orders belonging to the Office Supplies category. It helps understand the performance of this product category.

SQL Query :-

```
-- data of year 2017
SELECT *
FROM superstore
WHERE STR_TO_DATE(Order_Date, '%m/%d/%Y')
BETWEEN '2017-01-01' AND '2017-12-31';
```

Output :-

	Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_Name	Segment	Country	City	State
▶	13	CA-2017-114412	4/15/2017	4/20/2017	Standard Class	AA-10480	Andrew Allen	Consumer	United States	Concord	North Carolina
	24	US-2017-156909	7/16/2017	7/18/2017	Second Class	SF-20065	Sandra Flanagan	Consumer	United States	Philadelphia	Pennsylvania
	35	CA-2017-107727	10/19/2017	10/23/2017	Second Class	MA-17560	Matt Abelman	Home Office	United States	Houston	Texas
	42	CA-2017-120999	9/10/2017	9/15/2017	Standard Class	LC-16930	Linda Cazamias	Corporate	United States	Naperville	Illinois
	44	CA-2017-139619	9/19/2017	9/23/2017	Standard Class	ES-14080	Erin Smith	Corporate	United States	Melbourne	Florida
	72	CA-2017-114440	9/14/2017	9/17/2017	Second Class	TB-21520	Tracy Blumstein	Consumer	United States	Jackson	Michigan
	76	US-2017-118038	12/9/2017	12/11/2017	First Class	KB-16600	Ken Brennan	Corporate	United States	Houston	Texas
	77	US-2017-118038	12/9/2017	12/11/2017	First Class	KB-16600	Ken Brennan	Corporate	United States	Houston	Texas
	78	US-2017-118038	12/9/2017	12/11/2017	First Class	KB-16600	Ken Brennan	Corporate	United States	Houston	Texas

Insight :- This query retrieves all orders placed during 2017. It helps analyze sales for a specific year.

SQL Query :-

```
-- data where sales is between 500 and 1000
SELECT *
FROM superstore
WHERE Sales BETWEEN 500 AND 1000;
```

Output :-

	Row_ID	Postal_Code	Region	Product_ID	Category	Sub_Category	Product_Name	Sales	Quantity	Discount
▶	1	42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs,...	731.9400	3	0.00
	2	33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	957.5775	5	0.40
	3	90032	West	TEC-PH-10002275	Technology	Phones	Mitel 5320 IP Phone VoIP phone	907.1520	6	0.20
	4	90032	West	TEC-PH-10002033	Technology	Phones	Konftel 250 Conference phone - Charcoal black	911.4240	4	0.20
	5	53711	Central	OFF-ST-10004186	Office Supplies	Storage	Stur-D-Stor Shelving, Vertical 5-Shelf: 72"H x 3...	665.8800	6	0.00
	6	77041	Central	FUR-BO-10002545	Furniture	Bookcases	Atlantic Metals Mobile 3-Shelf Bookcases, Custo...	532.3992	3	0.30
	7	38109	South	FUR-CH-10000513	Furniture	Chairs	High-Back Leather Manager's Chair	831.9360	8	0.20
	8	98103	West	FUR-TA-10001768	Furniture	Tables	Hon Racetrack Conference Tables	787.5300	3	0.00
	9	77041	Central	FUR-CH-10004063	Furniture	Chairs	Global Deluxe High-Back Manager's Chair	600.5580	3	0.30

Insight :- This query shows medium-value sales transactions. It helps identify products and customers with moderate sales amounts.

SQL Query :-

```
-- data of central region and technology category
SELECT *
FROM superstore
WHERE Region = 'Central' AND Category = 'Technology';
```

Output :-

	Customer_Name	Segment	Country	City	State	Postal_Code	Region	Product_ID	Category	Sub_Category	Product_Name
▶	Gene Hale	Corporate	United States	Richardson	Texas	75080	Central	TEC-PH-10004977	Technology	Phones	GE 30524EE4
	Steve Nguyen	Home Office	United States	Houston	Texas	77041	Central	TEC-PH-10000486	Technology	Phones	Plantronics H
	Linda Cazamias	Corporate	United States	Naperville	Illinois	60540	Central	TEC-PH-10004093	Technology	Phones	Panasonic Kx
	Odella Nelson	Corporate	United States	Eagan	Minnesota	55122	Central	TEC-AC-10000171	Technology	Accessories	Verbatim 25 G
	Paul Gonzalez	Consumer	United States	Rochester	Minnesota	55901	Central	TEC-AC-10001998	Technology	Accessories	LogitechÂ L5:
	Rick Bensley	Home Office	United States	Chicago	Illinois	60610	Central	TEC-AC-10001767	Technology	Accessories	SanDisk Ultra
	Pete Armstrong	Home Office	United States	Orland Park	Illinois	60462	Central	TEC-AC-10000844	Technology	Accessories	LogitechÂ Ga
	Justin Ellison	Corporate	United States	Franklin	Wisconsin	53132	Central	TEC-PH-10000215	Technology	Phones	Plantronics C
	Justin Ellison	Corporate	United States	Franklin	Wisconsin	53132	Central	TEC-PH-10001448	Technology	Phones	Anker Astro :

Insight :- This query filters technology products sold in the Central region. It provides analysis for a specific category and region.

SQL Query :-

```
-- data of west or east region
SELECT *
FROM superstore
WHERE Region = 'West' OR Region = 'East';
```

Output :-

	Customer_ID	Customer_Name	Segment	Country	City	State	Postal_Code	Region	Product_ID	Category	Sub_Category
lass	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	OFF-BI-10002215	Office Supplies	Bin
lass	SF-20065	Sandra Flanagan	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-CH-10002774	Furniture	Chi
Class	EB-13870	Emily Burns	Consumer	United States	Orem	Utah	84057	West	FUR-TA-10000577	Furniture	Ta
lass	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	OFF-BI-10001634	Office Supplies	Bin
lass	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	TEC-AC-10003027	Technology	Acc
Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-BO-10004834	Furniture	Bo
Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	OFF-BI-10000474	Office Supplies	Bin
Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-FU-10004848	Furniture	Fur
Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	OFF-EN-10001509	Office Supplies	En

Insight :- This query combines records from both West and East regions. It helps compare sales activity across these regions.

SQL Query :-

```
-- data of west region and technology category where sales is greater than 1000
SELECT *
FROM superstore
WHERE Region = 'West' AND Category = 'Technology' AND Sales > 1000;
```

Output :-

	Postal_Code	Region	Product_ID	Category	Sub_Category	Product_Name	Sales	Quantity	Discount
nia	92037	West	TEC-AC-10003832	Technology	Accessories	Logitech P710e Mobile Speakerphone	3347.3700	13	0.0000
ia	85705	West	TEC-AC-10003911	Technology	Accessories	NETGEAR AC1750 Dual Band Gigabit Smart Wi...	1023.9360	8	0.2000
ngton	98198	West	TEC-CO-10002313	Technology	Copiers	Canon PC1080F Personal Copier	1799.9700	3	0.0000
nia	94110	West	TEC-CO-10004115	Technology	Copiers	Sharp AL-1530CS Digital Copier	1199.9760	3	0.2000
nia	59405	West	TEC-CO-10003236	Technology	Copiers	Canon Image Class D660 Copier	2999.9500	5	0.0000
nia	92024	West	TEC-CO-10004115	Technology	Copiers	Sharp AL-1530CS Digital Copier	1199.9760	3	0.2000
nia	94110	West	TEC-CO-10002095	Technology	Copiers	Hewlett Packard 610 Color Digital Copier / Printer	1199.9760	3	0.2000
ado	80219	West	TEC-PH-10001795	Technology	Phones	ClearOne CHATAttach 160 -A speaker phone	1983.9680	4	0.2000
nia	92704	West	TEC-AC-10001838	Technology	Accessories	Razer Tiamat Over Ear 7.1 Surround Sound PC ...	1399.9300	7	0.0000

Insight :- This query highlights high-value technology orders from the West region. These orders contribute significantly to revenue.

STEP 4. Aggregations using Group By Filters

SQL Query :-

```
-- Total Sales
SELECT Category, SUM(Sales) AS Total_Sales
FROM superstore
GROUP BY Category;
```

Output :-

Result Grid   Filter Rows:		
	Category	Total_Sales
▶	Furniture	741999.7953
	Office Supplies	719047.0320
	Technology	836154.0330

Insight :- This query shows how much revenue each category generates. It helps identify the best performing category.

SQL Query :-

```
-- Total Quantity
SELECT Category, SUM(Quantity) AS Total_Quantity
FROM superstore
GROUP BY Category;
```

Output :-

Result Grid   Filter Rows:		
	Category	Total_Quantity
▶	Furniture	8028
	Office Supplies	22906
	Technology	6939

Insight :- This query shows the number of items sold in each category. It helps understand customer demand.

SQL Query :-

```
-- Average Sales
SELECT Category, AVG(Sales) AS Avg_Sales
FROM superstore
GROUP BY Category;
```

Output :-

Result Grid			Filter Rows:
	Category	Avg_Sales	
▶	Furniture	349.83488699	
	Office Supplies	119.32410090	
	Technology	452.70927612	

Insight :- This query calculates the average sales amount for each category. It helps compare category performance.

SQL Query :-

```
-- Category wise total sales greater than 500000
SELECT Category, SUM(Sales) AS Total_Sales
FROM superstore
GROUP BY Category
HAVING SUM(Sales) > 500000;
```

Output :-

Result Grid			Filter Rows:	Export:
	Category	Total_Sales		
▶	Furniture	741999.7953		
	Office Supplies	719047.0320		
	Technology	836154.0330		

Insight :- This query displays categories with very high total sales. These categories are major contributors to business revenue.

STEP 5. Sorting and Limiting

SQL Query :-

```
-- Top Products
SELECT Product_Name, SUM(Sales) AS Total_Sales
FROM superstore
GROUP BY Product_Name
ORDER BY Total_Sales DESC
LIMIT 10;
```

Output :-

Result Grid   Filter Rows: Export:  Wrap C		
	Product_Name	Total_Sales
▶	Canon imageCLASS 2200 Advanced Copier	61599.8240
	Fellowes PB500 Electric Punch Plastic Comb Bind...	27453.3840
	Cisco TelePresence System EX90 Videoconferen...	22638.4800
	HON 5400 Series Task Chairs for Big and Tall	21870.5760
	GBC DocuBind TL300 Electric Binding System	19823.4790
	GBC Ibimaster 500 Manual ProClick Binding System	19024.5000
	Hewlett Packard LaserJet 3310 Copier	18839.6860
	HP Designjet T520 Inkjet Large Format Printer -...	18374.8950
	GBC DocuBind P400 Electric Binding System	17965.0680
	High Speed Automatic Electric Letter Opener	17030.3120

Result 25 ✕

Insight :- This query identifies the products with the highest sales. These products are important revenue contributors.

SQL Query :-

```
-- Top Categories
SELECT Category, SUM(Sales) AS Total_Sales
FROM superstore
GROUP BY Category
ORDER BY Total_Sales DESC;
```

Output :-

Result Grid   Filter Rows:		
	Category	Total_Sales
▶	Technology	836154.0330
	Furniture	741999.7953
	Office Supplies	719047.0320

Insight :- This query ranks categories based on total sales. It helps determine which category performs best.

STEP 6. Use Cases

SQL Query :-

```
-- Top Customers
SELECT Customer_Name, SUM(Sales) AS Total_Sales
FROM superstore
GROUP BY Customer_Name
ORDER BY Total_Sales DESC
LIMIT 10;
```

Output :-

Result Grid   Filter Rows:		
	Customer_Name	Total_Sales
▶	Sean Miller	25043.0500
	Tamara Chand	19052.2180
	Raymond Buch	15117.3390
	Tom Ashbrook	14595.6200
	Adrian Barton	14473.5710
	Ken Lonsdale	14175.2290
	Sanjit Chand	14142.3340
	Hunter Lopez	12873.2980
	Sanjit Engle	12209.4380
	Christopher Conant	12129.0720

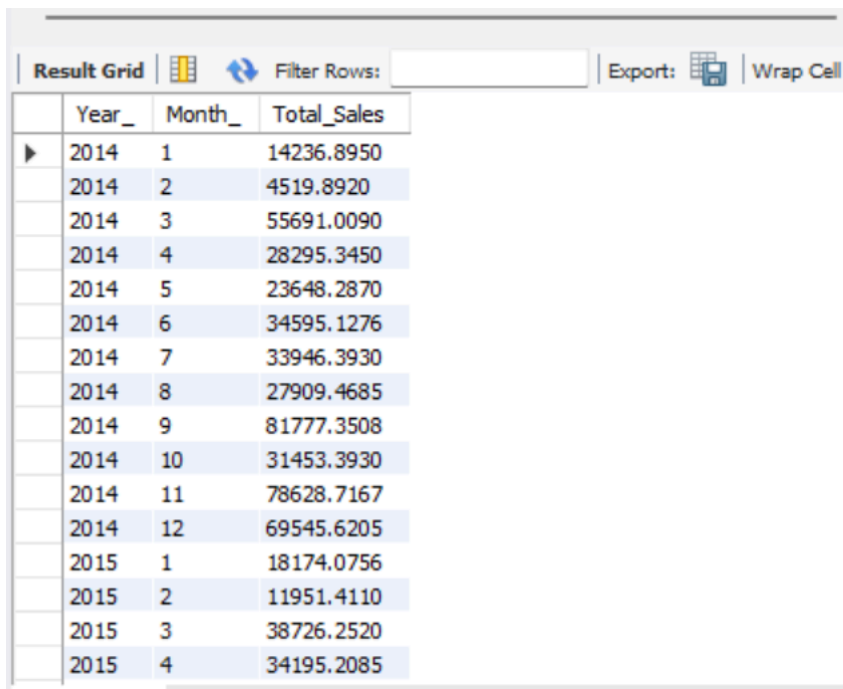
Result 27 

Insight :- This query identifies customers who generated the highest sales. These customers are valuable to the business.

SQL Query :-

```
-- Monthly Trend Sales
SELECT
    YEAR(STR_TO_DATE(Order_Date, '%m/%d/%Y')) AS Year_,
    MONTH(STR_TO_DATE(Order_Date, '%m/%d/%Y')) AS Month_,
    SUM(Sales) AS Total_Sales
FROM superstore
GROUP BY Year_, Month_
ORDER BY Year_, Month_;
```

Output :-



The screenshot shows a software interface with a 'Result Grid' tab. Above the grid are icons for a grid, a refresh arrow, a 'Filter Rows' input field, an 'Export' button with a document icon, and a 'Wrap Cell' button. The grid itself has three columns: 'Year_', 'Month_', and 'Total_Sales'. It contains 16 rows of data, representing the first four months of 2014 through 2015. The rows are highlighted in alternating light blue and white colors.

	Year_	Month_	Total_Sales
▶	2014	1	14236.8950
	2014	2	4519.8920
	2014	3	55691.0090
	2014	4	28295.3450
	2014	5	23648.2870
	2014	6	34595.1276
	2014	7	33946.3930
	2014	8	27909.4685
	2014	9	81777.3508
	2014	10	31453.3930
	2014	11	78628.7167
	2014	12	69545.6205
	2015	1	18174.0756
	2015	2	11951.4110
	2015	3	38726.2520
	2015	4	34195.2085

Insight :- This query shows how sales change over different months. It helps identify growth patterns and seasonal trends.

SQL Query :-

```
-- Duplicate Orders
SELECT Order_ID, COUNT(*) AS Cnt
FROM superstore
GROUP BY Order_ID
HAVING COUNT(*) > 1;
```

Output :-

Result Grid			Filter Rows:
	Order_ID	Cnt	
▶	CA-2016-152156	2	
	US-2015-108966	2	
	CA-2014-115812	7	
	US-2015-118983	2	
	CA-2014-143336	3	
	CA-2016-137330	2	
	CA-2016-121755	2	
	US-2015-150630	7	
	CA-2016-117590	2	
	CA-2015-117415	4	
	CA-2016-118255	2	

Insight :- This query finds Order IDs that appear multiple times. Multiple entries usually mean one order contains several products.

STEP 7. Data Validation

SQL Query :-

```
SELECT COUNT(*) AS Total_Rows
FROM superstore;
```

Output :-

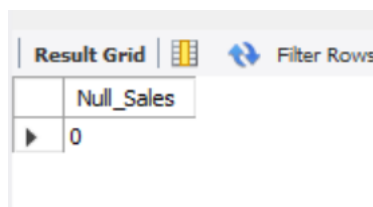
Result Grid	
	Total_Rows
▶	9994

Insight :- This query confirms the total number of records in the dataset. It helps verify that the data was imported successfully.

SQL Query :-

```
SELECT COUNT(*) AS Null_Sales
FROM superstore
WHERE Sales IS NULL;
```

Output :-



The screenshot shows a 'Result Grid' with a 'Filter Rows' button. The grid contains one column labeled 'Null_Sales' and one row with the value '0'.

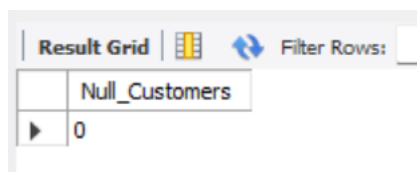
Null_Sales
0

Insight :- This query checks whether any sales values are missing. Missing values can affect analysis results.

SQL Query :-

```
SELECT COUNT(*) AS Null_Customers
FROM superstore
WHERE Customer_Name IS NULL;
```

Output :-



The screenshot shows a 'Result Grid' with a 'Filter Rows' button. The grid contains one column labeled 'Null_Customers' and one row with the value '0'.

Null_Customers
0

Insight :- This query checks whether any customer names are missing. Complete customer information improves reporting accuracy.

CONCLUSION

The dataset was analyzed using SQL queries for filtering, aggregation, sorting, and validation. The analysis provided insights into sales performance, customer behavior, product trends, and overall data quality.